

ECE4132 Assignment 2:

Modeling the temperature control lab system

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Aims

The aims of this assignment are

1. to gain familiarity with the simulation of systems defined by transfer functions in both MATLAB and Simulink
2. to fit the parameters of a simple dynamic model to noisy system data graphically and using an optimization-based approach.

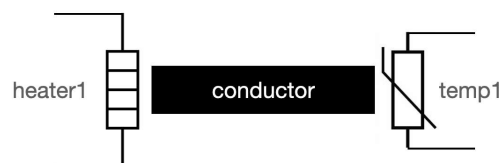
References and resources

- ‘Getting started with Simulink’ videos from Mathworks: <https://au.mathworks.com/videos/series/getting-started-with-simulink.html>
- N. S. Nise, *Control Systems Engineering*, 8th edition, Wiley.
- K. J. Åström and R. M. Murray *Feedback systems: an introduction for scientists and engineers* Princeton University Press

Introduction

In this assignment, we will be working with a digital twin of the temperature control system from lab 1. Throughout, we will call this the TClab system. In this lab we will only consider one heater input and one temperature output. The heater input is a value between 0 (no input) and 100 (maximum power). The temperature output is in degrees C. The temperature values are quantized and noisy.

A schematic of the system (only showing one heater and one temperature sensor) is shown below.



Note that the heater and the temperature sensor are physically separated by a conducting medium.

The aim of the assignment is to practice using methods to fit the parameters of different dynamic models of the temperature control system. These methods will each involve:

- Choosing a particular parametric form for the dynamic model (e.g., first order)
- Measuring the step response of the real system.
- Determining which choice of model parameters cause the step response predicted by the model to be as similar as possible to the measured step response.

The last step, of fitting model parameters, will be carried out in two different ways.

- The first method is ‘graphical’, based on looking for key features of the plots (e.g., rise times, etc) and choosing parameters accordingly.

- The second method will use simple numerical optimization tools to automate the process of finding a model that fits the data well.

Note: MATLAB/Simulink has powerful built-in system identification tools. We will not be using these, because we want you to get some understanding of what they are doing!

The temperature control system you will model is, in fact, most accurately modelled by a (mildly) nonlinear differential equation. However, in this assignment, all of the models you will come up with for the system will be linear (delay-)differential equation models defined by transfer functions.

Assessment

This assignment is worth 2% of the final grade for ECE4132. Submit your answers via the Moodle quiz marked **assignment 2**.

1 Transfer function models for the TClab system

Consider the TClab system which has a heater input and a temperature sensor output. Let

- $h(t)$ be the heater input at time t ,
- $T(t)$ be the temperature sensor output at time t ; and
- T_{room} be the room temperature.
- $D(t)$ be the difference between the output of the temperature sensor and room temperature, i.e.,

$$D(t) = T(t) - T_{\text{room}}. \quad (1)$$

Note: we change variables to work with $D(t)$ (the difference between the measured temperature and the ambient room temperature), because in these variables the system model can be modelled by a linear system. Indeed, when the input is $h(t) = 0$ for all time, then the temperature sensor should remain reading room temperature, and so $D(t) = 0$ for all time.

You will consider two different (simplified) models of the temperature control system in this lab, a first-order model and a first-order model with delay.

- **First-order model:** This models the relationship between h and D via the differential equation

$$\frac{dD}{dt} = -a_1 D(t) + K_1 h(t) \quad (2)$$

where a_1 and K_1 are model parameters. For this model, the temperature deviation when the heater input is a **unit** step is

$$D(t) = (K_1/a_1)(1 - e^{-a_1 t}).$$

Answer the following questions in the assignment 2 Moodle quiz:

- Q1 Find the transfer function from h to D for this first-order model of the system. Express your answer in terms of K_1 and a_1 and s .
- Q2 If $h(t) = 50 \times \mathbf{1}(t)$ is the unit step scaled by 50, and h and D are related by a first-order model, write down an expression for the steady-state value of $D(t)$. Express your answer in terms of K_1 and a_1 .

- **First-order model with delay:** This models the relationship between h and D via the delay-differential equation

$$\frac{dD}{dt} = -a_2 D(t) + K_2 h(t - \tau) \quad (3)$$

where a_2 and K_2 and τ are model parameters. This model explicitly captures the observation that there is a time delay between when the heater turns on and when the temperature sensor detects a change in temperature, because heat needs to conduct from one end of the conducting medium to the other. For this model, the temperature deviation when the heater input is a **unit** step is

$$D(t) = \begin{cases} (K_2/a_2)(1 - e^{-a_2(t-\tau)}) & \text{if } t \geq \tau \\ 0 & \text{otherwise.} \end{cases}$$

Answer the following questions in the assignment 2 Moodle quiz:

- Q3 Find the transfer function from h to D for this first-order model with delay of the system. Express your answer in terms of K_2 , a_2 , τ , and s .
- Q4 Suppose that $h(t) = 50 \times \mathbf{1}(t)$ is the unit step scaled by 50, and h and D are related by a first-order model with delay (with parameters K_2 , a_2 , and τ). If $T(t) = D(t) + T_0$ (where T_0 is the room temperature), write down an expression for the steady-state value of $T(t)$. Express your answer in terms of K_2 , a_2 , and T_0 .

2 System ID by manually fitting model parameters

2.1 Fitting model parameters from step response features

In this section you will measure the step response of the TClab system and save it.

- Download the `studentfiles.zip` file for assignment 2 from the Moodle page and unzip it.
- Open the Simulink file `TClabsim.slx`
- Add a 'step' source that steps from 0 to 50 at time $t = 0$. **Set the 'sample time' to 1.** Connect it to the Heater input.
- Save the system input to a file by sending it to the appropriate 'to file' block. Similarly save the system output to the appropriate 'to file' block.
- Add a 'scope' block and connect it to the temperature output (so you can monitor progress of your simulation).
- Simulate the system for 600 seconds.
- Once the simulation is complete, there should be files `Tsys.mat` and `usys.mat` in your working directory. If, for example, you `load Tsys.mat` there should be a variable `Tsys` in your workspace. Check that `Tsys.time` is a vector of time instants and that `Tsys.data` is a vector of signal values. Plot the response of the system.

(Optional) The following questions in the six bullerts below are included to guide you to measure various parameters of the step response that should help you answer Q5 and Q6. (These are not to be submitted in the Moodle quiz. They are just useful values to measure on your plot to help determine model parameters.)

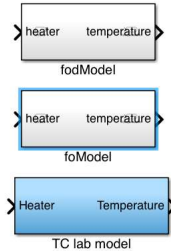
- What is the initial value of the temperature (this is the same as T_{room})?
- Approximately how long does the temperature stay at its initial value before it starts to rise? (i.e., how much delay is there in the system.)
- What is the steady-state value of the temperature?
- What is the steady-state value of the temperature *deviation* $D(t) = T(t) - T_{\text{room}}$?
- How long does it take from time 0 until the temperature deviation reaches 63% of its steady-state value?
- How long does it take from the time when the temperature starts to rise to the time the temperature deviation reaches 63% of its steady-state value?

Answer the following questions in the assignment 2 Moodle quiz:

- Q5 Find values for K_1 , a_1 and T_{room} such that the system response is well modelled by a first-order model (2) with these parameters.
- Q6 Find values for K_2 , a_2 , T_{room} , and τ such that the system response is well modelled by a first-order model with delay (3) with these parameters.

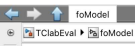
2.2 Model evaluation

In this section, you will use Simulink to simulate the first-order, and first-order with delay, models for the temperature control system using the parameter values you estimated in the previous section. You will compare the outputs with the response of the TClab system graphically. You will then see how your models compare with the response of the TClab system when *new input signals* are used (not just a scaled unit step input).



Begin by opening the Simulink model `TClabEval.slx`. This model has the TClab system, together with two subsystems which you will build your two models for the system.

2.2.1 Building the first-order model

- Double click on the `foModel` subsystem to edit it.
- Build a model of the first order system described by (2). Use the **Transfer Fcn** block together with the parameter values a_1 and K_1 you estimated in the previous section of the lab. (The transfer function you found in Q1 should help.)
- Add a constant to the output of your transfer function so that the output of the subsystem represents the temperature T rather than the temperature deviation D .
- Once you are done, go up to the parent model by clicking on the up arrow .

2.2.2 Building the first-order model with delay

You will now repeat what you did in the previous subsection for the first-order model with delay.

- Double click on the `fodModel` subsystem to edit it.
- Build a model of the first order system with delay described by (3). Use the **Transfer Fcn** block and the **Transport Delay** block together with the parameter values a_2 and K_2 and τ you estimated in the previous section of the lab. (The transfer function you found in Q3 should help.)
- Add a constant to the output of your transfer function so that the subsystem output represents the temperature T rather than the temperature deviation D .
- Once you are done, go up to the parent model.

2.2.3 Model evaluation

Connect up the subsystems in the parent model as shown below. Note that the sinks are ‘to workspace’ blocks.

3 System ID using simple numerical optimisation tools

The second part of the assignment involves developing methods to fit the parameters of the dynamical models you are considering (the first-order and first-order with delay models) using numerical optimisation procedures.

To do this, you will first need a function that evaluates the error between the (scaled) step response of the system (which you found and saved in section 1) and the (scaled) step response of a model of the system with particular choices of parameters. To construct such a function, we need to be able to compute the step response of a model given by a transfer function. While we could do this in Simulink, for simple models it is often faster to do this directly in MATLAB. Section 3.1 is devoted to simulating the first-order and first-order with delay models in MATLAB, and writing a function to compute the error between the model step response and the true system step response.

Given such a function, we can then call a numerical optimisation routine to try to adjust the parameters so as to minimise the error between the system step response and the model step response.

3.1 MATLAB simulation of a first-order model (with delay)

In this section you will simulate the first-order model, and the first-order model with delay, using MATLAB functions rather than Simulink. You will also compute the mean-squared error between the measured response of the TClab and the simulated response of the first-order model and the first-order with delay model. In the next section you will make use of these simulations to find the model parameters that minimise the mean-squared error.

Suppose that T_{sys} is the measured response of the TClab system to some input, sampled with sampling period 1 second. Similarly suppose that T_{model} is the corresponding simulated response of a system model to the same input, also sampled with sampling period 1 second. The mean squared error between the two responses is

$$\text{MSE} = \frac{1}{N+1} \sum_{n=0}^N (T_{sys}[n] - T_{model}[n])^2. \quad (4)$$

Note that the sum is from $n = 0$ to $n = N$, so the signals T_{sys} and T_{model} have length $N + 1$.

- Open and complete the MATLAB function `foCost.m` by uncommenting the indicated lines and filling in the parts labeled as ‘...’.

Answer the following questions in the assignment 2 Moodle quiz:

Q11 What value of the mean squared error do you obtain when you run `foCost` with parameters `K1= 0.002`, `a1= 0.02` and `Troom= 20`? (Note that these are not good choices of the parameters, we are just using them to test your function.) Round your answer correct to one decimal place.

- Open and complete the MATLAB function `fodCost.m` by uncommenting the indicated lines and filling in the parts labeled as ‘...’.

Answer the following questions in the assignment 2 Moodle quiz:

Q12 What value of the mean squared error do you obtain when you run `fodCost` with parameters `K2= 0.002`, `a2= 0.02`, `Troom= 20`, and `tau= 10`? (Note that these are not good choices of the parameters, we are just using them to test your function.) Round your answer correct to one decimal place.

3.2 Choosing optimal parameters

In this section of the lab you will use a simple numerical optimisation routine in MATLAB to find model parameters that minimize the mean squared error between the measured step response of the system and the step response of the model.

This optimisation method (the MATLAB function `fminsearch`) uses the *Nelder-Mead algorithm*, which is a very generic optimisation method. It has the advantage of only requiring access to a subroutine to evaluate the cost function at a given point. More sophisticated (and faster-converging) methods usually also require subroutines to compute the gradient and, in some cases, the Hessian of the cost function (assuming the cost function is sufficiently smooth).

- Open and complete the MATLAB script `optParamsFo.m` by uncommenting the indicated lines and filling in the parts labeled as ...
- Run the script and find the optimized values of the system parameters a_1 , K_1 and T_{room} .

Answer the following questions in the assignment 2 Moodle quiz:

Q13 What are the values of a_1 , K_1 , and T_{room} found when you run `optParamsFo`? Give all your answers correct to two significant figures.

- Open and complete the MATLAB script `optParamsFod.m` by uncommenting the indicated lines and filling in the parts labeled as ...
- Run the script and find the optimized values of the system parameters a_2 , K_2 , τ , and T_{room} .

Answer the following questions in the assignment 2 Moodle quiz:

Q14 What are the values of a_2 , K_2 , T_{room} , and τ found when you run `optParamsFod`? Give all your answers correct to two significant figures.

Q15 When searching for values of the model parameters that best fit the measured step response of the system, we did not add any constraints on the parameters. If we only wanted to search over stable models, what constraint must the parameters a_1 and a_2 satisfy?

- A $a_1 < 0$ and $a_2 < 0$
- B $a_1 < 0$ and $a_2 > 0$
- C $a_1 > 0$ and $a_2 < 0$
- D $a_1 > 0$ and $a_2 > 0$
- E $\text{Re}[a_1] < 0$ and $\text{Re}[a_2] < 0$
- F $|a_1| < 1$ and $|a_2| < 1$
- G No additional conditions, the models will be stable for any choice of a_1 and a_2
- H It is not enough to constrain a_1 and a_2 only to ensure stability of the resulting models. We also need to add constraints on the other parameters (such as K_1 and K_2 , or τ or T_{room}).